Multiplying by One Digit

Break the big number into tens and ones. Multiply each part. Add.

23 × 4: break 23 into 20 and 3



$4 \times 20 = 80$

$4 \times 3 = 12$

$80 + 12 = 92$

Three digits work the same way: 146 × 3



$300 + 120 + 18 = 438$

Hundreds, tens and ones.

No picture? Write the parts in a list

$5 \times 30 = 150$

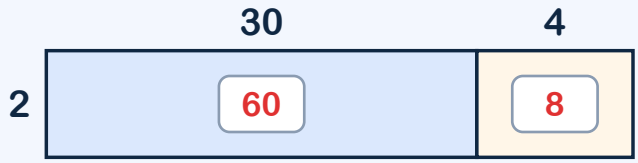
$5 \times 7 = 35$

$150 + 35 = 185$

37 × 5 = 185

Tens times 5, ones times 5, then add.

Your turn!



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For teachers

Goal: multiply a two- or three-digit number by one digit with the area model, then partial products (4.NBT.5). It is packet 39's break-it-apart area with packet 38's tens. Say “Tens times 4, ones times 4, then add.”

Multiplying by One Digit: Practice

Break it apart. Multiply each part. Add the parts.

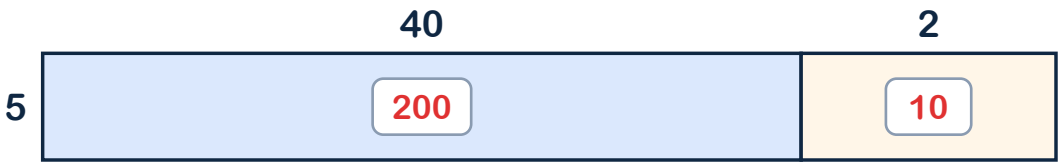
Part 1 The area model The parts are on top. Multiply each part.

1) $26 \times 3 = 78$



60 + 18 = 78

2) $42 \times 5 = 210$



200 + 10 = 210

3) $17 \times 6 = 102$



60 + 42 = 102

Multiplying by One Digit: Practice (continued)

Part 1 The area model (continued)

4) $35 \times 4 = 140$



120 + 20 = 140

Part 2 Split it yourself Write the tens and ones on top first.

5) $48 \times 2 = 96$



80 + 16 = 96

6) $63 \times 7 = 441$



420 + 21 = 441

Multiplying by One Digit: Practice (continued)

Part 2 Split it yourself (continued)

7) $29 \times 8 = 232$



160 + 72 = 232

8) $54 \times 6 = 324$



300 + 24 = 324

Part 3 Three digits Hundreds, tens and ones.

9) $213 \times 4 = 852$

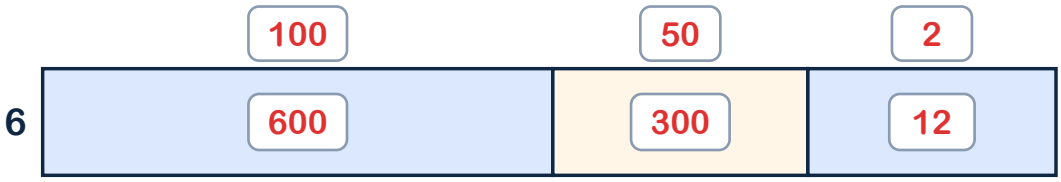


800 + 40 + 12 = 852

Multiplying by One Digit: Practice (continued)

Part 3 Three digits (continued)

10) $152 \times 6 = 912$



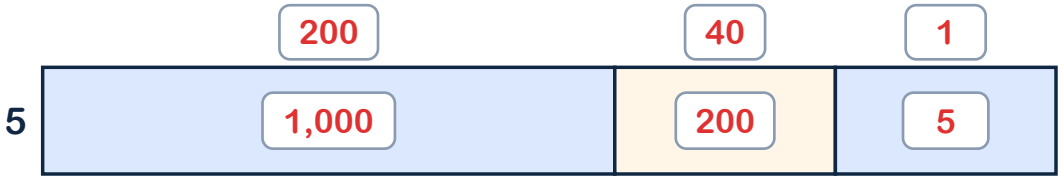
600 + 300 + 12 = 912

11) $324 \times 3 = 972$



900 + 60 + 12 = 972

12) $241 \times 5 = 1{,}205$



1,000 + 200 + 5 = 1,205

Multiplying by One Digit: Practice (continued)

Part 4 Write the parts No picture this time.

13) 36×4

$4 \times 30 = 120$

$4 \times 6 = 24$

$120 + 24 = 144$

14) 71×3

$3 \times 70 = 210$

$3 \times 1 = 3$

$210 + 3 = 213$

15) 58×6

$6 \times 50 = 300$

$6 \times 8 = 48$

$300 + 48 = 348$

16) 124×7

$7 \times 100 = 700$

$7 \times 20 = 140$

$7 \times 4 = 28$

$700 + 140 + 28 = 868$

Part 5 Two digits No help this time.

17) $32 \times 4 = 128$

18) $57 \times 3 = 171$

19) $46 \times 5 = 230$

20) $83 \times 2 = 166$

21) $19 \times 7 = 133$

22) $64 \times 3 = 192$

23) $38 \times 6 = 228$

24) $75 \times 4 = 300$

25) $91 \times 8 = 728$

26) $27 \times 9 = 243$

27) $44 \times 7 = 308$

28) $66 \times 5 = 330$

Part 6 Three digits No help this time.

29) $214 \times 3 = 642$

30) $132 \times 4 = 528$

Multiplying by One Digit: Practice (continued)

Part 6 Three digits (continued)

31) 305 × 6 = 1,830

32) 421 × 5 = 2,105

33) 163 × 2 = 326

34) 247 × 3 = 741

35) 512 × 7 = 3,584

36) 138 × 4 = 552

Part 7 Remember? Problems from earlier packets.

37) 3,846 to the nearest hundred = 3,800

38) 2,457 + 1,386 = 3,843

39) in 7,062 the 7 is worth 7,000

40) 4 × 20 = 80

41) 6 L + 9 L = 15 L

42) 5,000 − 1,246 = 3,754

Part 8 Word problems Draw the area model if it helps.

43) A box holds 24 crayons. How many crayons are in 6 boxes?

Number sentence 24 × 6 = 144

Answer 144 crayons

Draw or show your work

44) A bus has 48 seats. How many seats are on 4 buses?

Number sentence 48 × 4 = 192

Answer 192 seats

Draw or show your work

Multiplying by One Digit: Practice (continued)

Part 8 Word problems (continued)

45) Tickets cost $35 each. How much do 4 tickets cost?

Number sentence 35 × 4 = 140

Answer $140

Draw or show your work

46) A farmer plants 7 rows of 126 corn plants. How many plants is that?

Number sentence 126 × 7 = 882

Answer 882 plants

Draw or show your work

47) Challenge: A school has 5 classes of 28 kids and 4 classes of 31 kids. How many kids in all?

Number sentence 28 × 5 = 140, 31 × 4 = 124, 140 + 124 = 264

Answer 264 kids

Draw or show your work

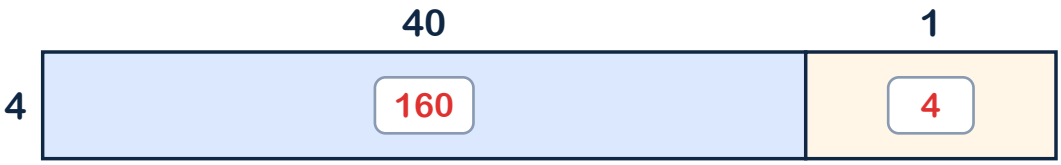
Check-In: Multiplying by One Digit

Show what you know. No help this time.

Mastery: 7 of 8 (80%)

Part 1 The area model

1) $41 \times 4 =$ 164



160 + 4 = 164

Part 2 Write the parts

2) 67×4

$4 \times 60 =$ 240

$4 \times 7 =$ 28

240 + 28 = 268

3) 89×5

$5 \times 80 =$ 400

$5 \times 9 =$ 45

400 + 45 = 445

Part 3 Multiply

4) $29 \times 6 =$ 174

5) $45 \times 5 =$ 225

6) $96 \times 9 =$ 864

Part 4 Three digits

7) $431 \times 4 =$ 1,724

8) $199 \times 7 =$ 1,393